

KIRILLOVKA, Ukraine

WMO: 346090

Lat: 47.3297N

Lon: 36.3350E

Elev: 725

StdP: 14.31

Time Zone: 2.00 (EUE)

Period: 86-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.4	(c) 5.9	(d) -2.8	(e) 4.9	(f) 2.7	(g) 1.4	(h) 6.1	(i) 6.7	(j) 39.0	(k) 26.5	(l) 37.2	(m) 28.0	(n) 3.5	(o) 50	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	88.5	68.7	85.6	67.6	82.6	66.7	71.1	83.5	69.6	81.5	68.1	79.8	5.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	102.9	76.3	65.5	96.7	74.8	63.9	91.2	73.8	35.5	83.3	34.1	81.1	32.8	79.9	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 36.4	(b) 33.1	(c) 25.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 36.4	(b) 33.1	(c) 25.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	26.7	27.8	37.0	48.3	58.6	66.3	69.8	69.6	60.7	48.5	35.9	28.3
	DBStd	17.90	9.82	9.92	8.13	7.51	6.92	5.39	5.43	5.72	7.76	8.88	10.18	9.04
	HDD50	3128	723	621	409	118	12	0	0	0	8	137	427	673
	HDD65	6602	1188	1040	869	500	218	51	17	23	172	513	873	1138
	CDD50	2487	0	0	5	68	280	488	615	607	329	90	5	0
	CDD65	486	0	0	0	0	21	89	168	164	43	1	0	0
	CDH74	4519	0	0	0	1	177	690	1617	1657	372	5	0	0
	CDH80	1326	0	0	0	0	14	139	502	565	106	0	0	0
Wind	WSAvg	6.8	7.5	6.8	8.4	7.6	6.8	6.4	5.8	5.2	5.7	8.4	6.0	7.0
Precipitation	PrecAvg	22.2	1.7	1.4	1.7	1.8	1.7	2.8	2.2	1.9	1.5	1.4	1.8	1.8
	PrecMax	37.5	6.0	3.2	3.5	4.1	4.1	8.3	5.9	6.4	5.6	3.4	4.9	4.7
	PrecMin	14.8	0.8	0.2	0.7	0.0	0.0	0.7	0.0	0.1	0.0	0.2	0.0	0.3
	PrecStd	5.3	1.1	0.8	0.8	1.1	1.1	1.8	1.8	1.9	1.3	0.8	1.2	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	54.1	64.4	73.1	81.8	86.6	91.4	92.9	89.2	74.9	58.6	51.9
		MCWB	43.6	48.8	50.1	56.9	64.2	67.9	71.6	68.1	67.3	60.0	53.7	51.1
	2%	DB	42.6	45.5	58.2	68.6	78.5	83.6	88.1	89.1	82.8	70.2	55.3	45.2
		MCWB	41.7	43.2	48.4	55.1	62.7	67.0	69.4	68.1	64.7	58.1	51.9	43.8
	5%	DB	40.1	42.1	53.1	64.9	75.7	80.8	85.2	85.8	78.4	65.4	52.4	41.5
		MCWB	39.2	40.6	45.9	52.5	61.8	67.1	68.3	67.4	62.8	57.0	49.9	40.7
	10%	DB	37.6	39.4	48.6	61.7	71.9	77.9	82.0	82.6	73.9	61.2	48.9	39.0
		MCWB	36.8	38.0	43.1	51.5	59.5	65.5	66.7	67.0	60.8	55.1	46.9	38.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	49.5	52.2	58.3	69.5	71.6	73.4	73.7	69.7	62.9	55.7	51.3
		MCDB	44.9	53.3	62.0	69.1	79.4	80.7	86.0	83.4	85.3	69.7	57.6	52.1
	2%	WB	41.9	43.9	49.0	56.3	65.5	69.7	71.2	70.9	66.4	60.6	52.7	43.8
		MCDB	42.6	45.1	57.0	65.2	74.7	79.4	84.8	83.9	79.2	65.4	54.8	44.9
	5%	WB	39.0	41.1	46.5	54.2	63.1	68.2	69.6	69.0	64.1	58.2	50.2	40.7
		MCDB	39.7	42.3	52.5	62.9	71.5	77.2	82.2	81.9	75.0	63.7	51.8	41.3
	10%	WB	36.9	38.3	44.0	52.2	61.0	66.7	67.9	67.5	62.0	55.2	47.3	38.2
		MCDB	37.6	39.4	47.6	60.2	68.8	75.6	79.4	80.3	71.2	61.7	49.0	38.9
Mean Daily Temperature Range	5% DB	MDBR	8.5	11.2	13.6	17.4	18.8	17.8	20.9	22.0	18.8	16.4	10.7	8.1
		MCDBR	8.3	12.4	21.0	23.8	22.1	22.3	23.9	24.7	24.2	21.3	13.4	8.3
		MCWBR	8.0	10.4	13.0	13.0	10.6	9.9	9.9	11.0	11.6	11.9	10.3	7.7
	5% WB	MCDBR	8.2	12.2	18.9	20.0	18.0	19.2	21.8	23.3	22.4	17.1	10.7	8.1
		MCWBR	7.7	10.7	12.0	11.5	10.2	9.6	10.1	12.7	12.0	10.8	9.5	7.8
Clear-Sky Solar Irradiance	taub	0.330	0.360	0.390	0.444	0.409	0.410	0.434	0.432	0.404	0.371	0.347	0.332	
	taud	2.257	2.198	2.205	2.089	2.281	2.301	2.238	2.234	2.289	2.350	2.342	2.300	
	Ebn at Noon	230	248	260	256	272	271	262	257	253	243	224	215	
	Edh at Noon	25	33	38	48	41	40	42	41	35	29	23	22	
All-Sky Solar Radiation	RadAvg	310	598	949	1468	1906	2006	2003	1812	1289	771	371	240	
	RadStd	56	103	116	139	164	148	111	118	112	105	57	40	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+1.37	N/A	N/A	N/A	+2.12	N/A	N/A	-252	-335	+227	+153

Nomenclature: See separate page